Topic 1: Number

1.1 Integers

Subject content	Guidance	Curriculum reference
A Add, subtract, multiply and divide positive and negative numbers in context	The temperature in London is -3°C Paris is 7° warmer than London. Oslo is twice as cold as London. How much warmer is Paris than Oslo?	N7.1B N8.1A
B Use the terminology of primes, multiples, factors, common multiples, common factors, highest common factor and lowest common multiple	$16, 19, 47, 72, 96$ Which of the numbers above is a prime? Which of the numbers above is a multiple of 8? Which of the numbers above is a factor of 144?	N7.1D N7.1E N7.1G
C Identify common prime factors, common factors and common multiples	What are the common factors of 20 and 24? Which of these is a multiple of both 6 and 8? $12, 24, 32, 40, 64$	N7.1D N7.1E N7.1G
D Identify highest common factors (HCF), lowest common multiples (LCM) and prime factors and use to solve problems	What is the HCF of 20 and 24? What is the LCM of 20 and 24? Burgers come in packs of 20 Buns come in packs of 12 What is the smallest number of burgers and buns you can buy so that every burger has a bun?	N7.1D N7.1E N8.1E
E Express an integer as a product of its prime factors	Write 180 as a product of its prime factors	N7.1F N8.1D

1.1 Integers *continued*

Subject content		Guidance	Curriculum reference
F	Know square numbers up to 144	Which of these is a square number? $1, 16, 19, 64, 72$	N7.1H
G	Know cube numbers up to 125	Which of these is a cube number? $1, 16, 19, 64, 72$	N7.1I
H	Know exact square roots up to 144 and exact cube roots up to 125	$\sqrt{144}$ $\sqrt[3]{125}$	N7.1H N7.1I
I	Use index notation and index laws	Write $3 \times 3 \times 3 \times 3$ as a single power of 3 Write $4^5 \times 4^7$ as a single power of 4 Write $5^6 \div 5^4$ as a single power of 5 Write $(6^4)^5$ as a single power of 6	N7.1J N8.1C
J	Use index laws with zero and negative indices	Find the value of 153^0 Evaluate 64^{-3}	N9.1C
K	Use order of operations to calculate combinations of powers, roots and brackets	$5^2 + 3 - (6 + \sqrt{4} \times 3.5)$	N7.4B N8.1B
L	Round integers up to three significant figures	What is $104\,975$ to 3sf?	N9.1A N9.1B
M	Identify upper and lower bounds for discrete data	The length of a path is 47 m to the nearest metre. What is the minimum possible length of the path?	N9.1E